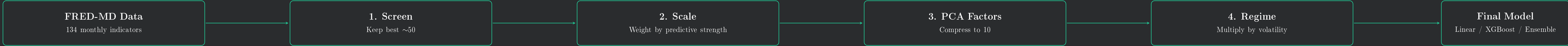


Can Economic Data Predict the Stock Market?

A Machine Learning Approach to S&P 500 Forecasting

SSRF Pipeline: From Economic Data to Market Prediction



What We Built

We built a pipeline that reads **130+ monthly economic statistics** (jobs, inflation, housing, interest rates) and predicts whether the S&P 500 will go up or down. We tested two horizons: next-month and next-quarter forecasts, simulating real-time trading over 26 years.

Data

Input: 134 monthly economic indicators from the Federal Reserve's FRED-MD database (1979–2026). Covers employment, industrial production, housing, retail sales, interest rates, inflation, and consumer sentiment.
Target: S&P 500 index returns from Yahoo Finance. Two targets: (1) next month's return, and (2) cumulative return over the next 3 months.
Size: 540 months, 134 features. Training starts with 120 months (10 years) and expands walk-forward — no future data leaks in.

Features

Raw indicators are noisy and highly correlated. We process them in three steps before feeding them to any model:

- Group-wise Screen** — Each indicator gets a t-statistic measuring how well it predicts returns on its own. Within each economic category, we keep only the ones that show some signal. For example, of 20 interest-rate indicators (3-month T-bill, 10-year yield, corporate spreads, etc.), only the strongest few survive. Same for jobs, inflation, housing — preventing any single category from flooding the model with noise. Typically 40–60 indicators remain.
- Predictive Scaling** — Each surviving indicator is multiplied by how strongly it has historically predicted returns. For example, if the unemployment rate has been a strong predictor, it gets amplified; if housing starts have been a weak one, it gets shrunk. This way the next step focuses on what actually matters for forecasting.
- Factor Extraction (PCA)** — The remaining indicators still move together (they all track the same economy). Principal Component Analysis compresses them into 10 uncorrelated “economic factors” — e.g., all employment indicators combine into a single labor factor, all interest rates into a rates factor, all inflation measures into a prices factor.

We also create **regime interaction terms** — multiplying each factor by recent market volatility — so the model adapts to calm vs. turbulent markets.

Models

Seven models tested on the same pipeline:

Model	Description
Elastic Net	Linear model with feature selection
Linear Regression	Simple linear combination of factors
XGBoost	Gradient-boosted decision trees
Random Forest	Ensemble of many decision trees
CatBoost	Advanced tree model
MLP (Neural Net)	2-layer neural network
Ensemble	Average of Linear + XGBoost

Baselines: Always-long (predict market always goes up) and momentum (predict same direction as last month).

Results

	1-Mo	3-Mo	3-Mo
Model	Hit Rate	Hit Rate	Sharpe
CatBoost	56.9%	74.5%	0.85
Ensemble	52.8%	70.3%	0.87
XGBoost	56.7%	66.2%	0.82
Linear	50.7%	65.5%	0.74
Random Forest	54.6%	62.8%	0.61
Elastic Net	55.0%	59.3%	0.33
MLP (Neural Net)	47.2%	62.1%	0.33
Always-Long (baseline)	62.5%	66.6%	—
Buy & Hold S&P 500	—	—	0.68

Key Takeaways

- Monthly:** No model beats always-long (62.5%)
- 3-month (overlapping):** CatBoost 74.5% beats baseline
- 3-month (non-overlap):** RF 72.3%, signal confirmed
- No model predicts *how much* market moves
- Signal exists at quarterly horizon, not monthly

Discussion

Monthly: All models fail. None beat the simple strategy of always predicting up (62.5% accuracy). This matches the well-known result that next-month stock returns are unpredictable from economic data.
Quarterly: Real signal appears. CatBoost (74.5%) beats the 66.6% baseline by nearly 8 points. The Ensemble reaches 70.3% with the best risk-adjusted return (Sharpe 0.87).
But the signal is weak. Models get *direction* right but can't predict *magnitude*. All have negative R^2_{OOS} , meaning a simple historical average predicts magnitudes better.
Non-overlapping evaluation confirms signal. Using independent quarterly observations (-non-overlap flag), Random Forest achieves 72.3% hit ratio and MLP a 0.84 Sharpe – confirming direction signal is not an artifact of overlapping windows.
Leverage helps manage risk. Asymmetric sizing (2.5× long, 0.25× short) cuts maximum loss from ~47% (buy & hold) to ~10%, though it slightly reduces Sharpe for the strongest models.

What Didn't Work

- Extreme leverage** (25×): destroyed all portfolios
- Monthly returns:** hopeless at this horizon
- Magnitude prediction:** all R^2_{OOS} negative
- Transaction costs:** eat 1-2% of annual returns

Future Work

- Model actual release lags of economic data (FRED reports come weeks after month-end)
- Replace volatility proxy with a Hidden Markov Model for better regime detection
- Full transaction cost modeling for production systems

References

- Goyal & Welch (2008). A Comprehensive Look.
- Campbell & Thompson (2008). Predicting Excess Returns.
- McCracken & Ng (2016). FRED-MD Database.
- Zou & Hastie (2005). Elastic Net.

Code: https://github.com/pellucide/sp500_macro_forecast-